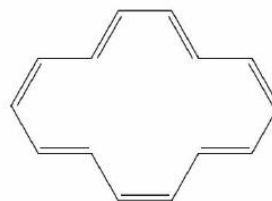
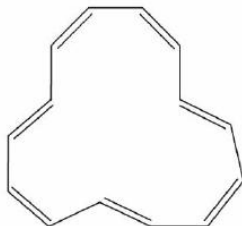


40. The ^1H NMR spectrum of naphthalene shows two multiplets (Figure 15-16). The upfield absorption ($\delta = 7.49 \text{ ppm}$) is due to the hydrogens at C2, C3, C6, and C7, and the downfield multiplet ($\delta = 7.86 \text{ ppm}$) is due to the hydrogens at C1, C4, C5, and C8. Explain why one set of hydrogens is deshielded more than the other.

64. **CHALLENGE** (a) The ^1H NMR spectrum of [18]annulene shows two signals, at $\delta = 9.28$ (12 H) and -2.99 (6 H) ppm. The negative chemical shift value refers to a resonance upfield (to the right) of $(\text{CH}_3)_4\text{Si}$. Explain this spectrum. (**Hint:** Consult [Figure 15-9](#).) (b) The unusual molecule 1,6-methano[10]annulene (shown below) exhibits two sets of signals in the ^1H NMR spectrum at $\delta = 7.10$ (8 H)

65. The ^1H NMR spectrum of the most stable isomer of [14]annulene shows two signals, at $\delta = -0.61$ (4 H) and 7.88 (10 H) ppm. Two possible structures for [14]annulene are shown here. How do they differ? Which one corresponds to the NMR spectrum described?



- 49.** The NMR and IR spectra for four unknown compounds A through D are presented below. Possible empirical formulas for them (not in any